[1, 3, 5, 11 → [$K^2/2$]

→ 2, 4, 7, 12

→ 1, 1, 3, 3 → [$K^2/2$ ~~$K^2 = K+1$~~]

13, 14, 15, 16]

O/P: 1, 1, 1, 2, 3, 3, 3, 4, 5, 7, 11, 12, 13, 14, 15, 16

BF.

[1, 3, 5, 11] → [1, 2, 3, 4, 5, 7, 11, 12]

TC: $2K + 3K + \dots + K^2$

$= K(2 + 3 + 4 + 5 \dots K)$

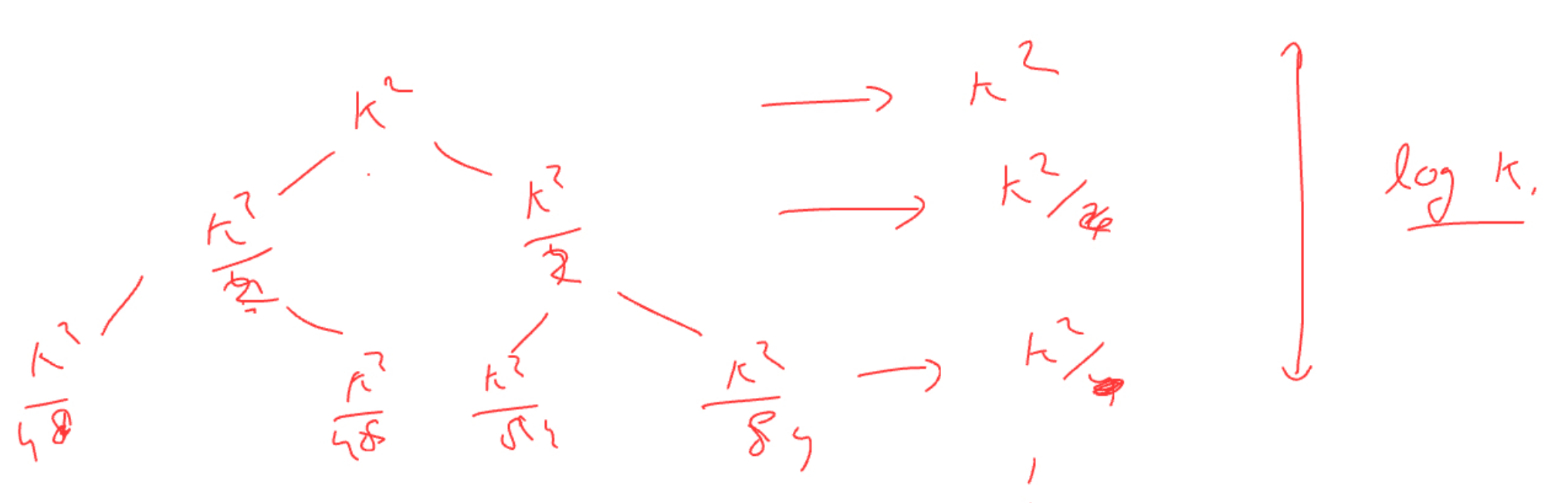
$= O(K^3)$

AS = $O(K^2)$.

Divide & Conquer.

$T(K, K)$

$T(K^2) = \underbrace{2T\left(\frac{K^2}{2}\right)}_{} + K^2$ TODO: Solve the recurrence.


K^2
$\frac{K^2}{2}$
$\frac{K^2}{2}$
$\frac{K^2}{4}$
$\frac{K^2}{4}$
$\frac{K^2}{4}$
$\frac{K^2}{4}$
K^2
$K^2/2$
$K^2/4$
log K.

TC = $O(K^2 \log K)$

i, j, j+1

i [1, 3, 5, 11

→ 2, 4, 7, 12

→ 1, 1, 3, 3

13, 14, 15, 16]

1, 2, 1, 13

{1

List → K < K^2

Min-Heap. (log K) * K^2 → $O(K^2 \log K)$

AS → $O(K)$.

↓

Min-Heap. { arr[i][j], [i, j] }